A Decision Variables Classification-Based Evolutionary Algorithm for Constrained Multi-Objective Optimization Problems

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***Abstract*—Solving constrained multi-objective optimization problems (CMOPs) is a challenging task due to the presence of multiple conflicting objectives and intricate constraints. In order to better address CMOPs and achieve a balance between objectives and constraints, existing constrained multi-objective evolutionary algorithms (CMOEAs) predominantly focus on devising various strategies by leveraging the relationships between objectives and constraints, and the designed strategies usually are effective for the problems with simple constraints. However, these methods most ignore the relationship between decision variables and constraints. In fact, the essence of optimization is to find appropriate decision variables to meet various complex constraints. Therefore, it is hoped that the problem can be analyzed from the perspective of decision variables, so as to obtain more excellent results. Based on the above motivation, this paper proposes a decision variables classification approach, according to the relationship between decision variables and constraints, variables are divided into constraint-related (CR) variables and constraint-independent (CI) variables. Consequently, by optimizing these two types of variables independently, the population can sustain a favorable balance between feasibility and diversity. Furthermore, specific offspring generation strategies are proposed for the two categories of decision variables in order to achieve rapid convergence while maintaining population diversity. Experimental results on 31 test problems as well as 20 real-world problems demonstrate that the proposed algorithm is competitive compared to some state-of-the-art constrained multi-objective optimization algorithms.**

***Index Terms*—Constraint-independent (CI), constrained multi-objective optimization, constraint-related (CR), decision variables.**

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I. Introduction

CONSTRAINED multi-objective optimization problems are a widespread class of problems in practical life, for example: when optimizing the trajectory of a spacecraft [1], the cross-distance ratio, aerodynamic total heat, heat flux density, and smoothness are used as optimization objectives, but they are also limited by factors such as heating rate, dynamic pressure, and acceleration. Another example is that in the overtaking process of autonomous vehicles [2], the overtaking duration, visibility, and comfort are taken as the objectives, but at the same time, constraints such as safety, road width, and obstacles are also taken into consideration. In addition, in the economic dispatch of the power system [3], fuel cost and pollution emission are optimization objectives, but they are also limited by generation capacity, generation power, and transmission power. This type of problem that considers multiple objectives and constraints simultaneously is called CMOPs (constrained multi-objective optimization problems), which widely exists in many fields such as industrial production [4], optimal scheduling [5], path planning [6], and energy optimization [7]. It is of great practical significance to solve such problems. Generally, a CMOP in the minimization case can be represented by the following formula [8]:

$$\min \vec{F}(\vec{x}) = (f_1(\vec{x}), f_2(\vec{x}), \ldots, f_m(\vec{x}))^T$$
$$\text{s.t.} \begin{cases} g_j(\vec{x}) \leq 0, j = 1, \ldots, l \\ h_j(\vec{x}) = 0, j = l+1, \ldots, q \\ \vec{x} = (x_1, x_2, \ldots, x_D)^T \in \mathbb{S} \end{cases} \tag{1}$$

where $\vec{F}(\vec{x})$ denotes the objective vector, typically encompasses m conflicting objectives that necessitate optimization. $g_j(\vec{x})$ and $h_j(\vec{x})$ represent the j-th inequality and $(j-l)$-th equation constraint that need to be satisfied, respectively. The symbols l and $(q$-$l)$ respectively represent the number of inequality and equality constraints. The solution vector, $\vec{x}$, contains D decision variables, and its search space is represented by $\mathbb{S}$.

For any individual $\vec{x}$, the degree to which it violates the j-th constraint can be represented as follows:

$$G_j(\vec{x}) = \begin{cases} \max\left(0, g_j(\vec{x})\right), & j = 1, \ldots, l \\ \max\left(0, \left|h_j(\vec{x})\right| - \delta\right), & j = l+1, \ldots, q \end{cases} \quad (2)$$

where the notation δ denotes a minute positive value that is employed to ease the strictness of the equality constraints. Consequently, the overall extent of constraint violation for the individual $\vec{x}$ can be determined by aggregating its violations across all constraints.

$$G(\vec{x}) = \sum_{j=1}^{q} G_j(\vec{x}). \quad (3)$$

When confronted with a CMOP, an individual is deemed to be a feasible solution if it satisfies all constraints, otherwise, it is considered an infeasible solution. In the case of two individuals, denoted as $\vec{x}_a$ and $\vec{x}_b$, solution $\vec{x}_a$ is said to Pareto dominate $\vec{x}_b$ if $f_n(\vec{x}_a) \leq f_n(\vec{x}_b)$ for all n in the set $\{1, 2, \ldots, m\}$, and $f_r(\vec{x}_a) < f_r(\vec{x}_b)$ for at least one r in the set $\{1, 2, \ldots, m\}$. The solution $\vec{x}_a$ is considered the Pareto optimal solution if it is not dominated by any alternative solution [9]. The set of Pareto optimal solutions consisting of feasible solutions is known as the constrained Pareto front (CPF), and the primary purpose of solving a CMOP is to locate a CPF that is well converged and uniformly distributed. Meanwhile, its counterpart is the unconstrained Pareto front (UPF) consists of the Pareto optimal solutions in the case where all constraints are ignored and the feasibility of solutions is not considered.

To effectively address CMOPs, numerous constrained multi-objective evolutionary algorithms (CMOEAs) are proposed, which can be roughly divided into three categories: 1) Methods based on single population and single stage [10]–[12], 2) Methods based on two-stage [13]–[15], and 3) Methods based on two-population [16]–[18]. These methods typically utilize the relationship between objectives and constraints to assist in the evolution of the population, thereby searching for CPF. However, they have certain limitations in solving problems. For the first category of algorithms, they often emphasize objectives optimization to increase population diversity in the early stage of evolution, while the satisfaction of constraints is focused during the later stage of evolution. However, such algorithms are more suitable for problems where the trends of objectives or constraints are relatively regular. As for the second and third categories of algorithms, they often leverage information from the UPF to aid in the search for the CPF, or investigate the correlation between UPF and CPF, and then design effective strategies. However, they can only achieve excellent results when the UPF and CPF are in the same search direction, or in cases where they are close to each other. Once the UPF is significantly far away from the CPF, or there is no positive relationship between the objectives and the constraints, the information from the UPF may not provide significant assistance in locating CPF, and even misleads the population to evolve in the wrong direction.

Based on the above analysis, these methods can achieve satisfactory results on problems with simple constraints, but they ignore the relationship between decision variables and constraints, once the constraints fitness landscape of the problem becomes complex, the performance will significantly degrade. In fact, the purpose of optimization is to find a suitable set of decision variables to satisfy various constraints, and the constraints exhibit a direct connection with the decision variables. And we have found that some practical problems, such as pressure vessel design [19]–[20], are not all variables related to constraints. Therefore, if the variables related to constraints can be identified, they can be optimized to help the population quickly locate the promising feasible regions. At the same time, the decision variables unrelated to constraints can be optimized to increase the diversity of the population. However, this phenomenon is ignored by most studies. Hence, this paper endeavors to investigate the intricate relationship between decision variables and constraints, aiming to facilitate the resolution of complex CMOPs. In this context, decision variables are categorized into distinct types based on their characteristics, subsequently undergoing separate optimization processes. The principal contributions of this paper are outlined as follows:

1) A decision variables classification-based evolutionary algorithm, named DVCEA, is proposed to solve CMOPs. In this method, a simple classification approach for decision variables is designed, specifically, the decision variables undergo perturbations to assess alterations in constraint violation degrees. In this way, decision variables are divided into two categories based on their relationship with constraints: constraint-related (CR) variables and constraint-independent (CI) variables. Afterwards, these two types of variables are separately optimized.

2) Due to the presence of multiple local infeasible regions in the fitness landscape of constraints for the majority of problems, the population may converge to a local region if the two kinds of decision variables are only optimized separately without applying targeted optimization strategies. Hence, two diverse variables optimization strategies are studied for different decision variables, with the aim of helping the population converge quickly while maintaining diversity.

In addition, the experimental results on 31 test problems with complex and simple relationships between objectives and constraints, along with 20 real-world problems, demonstrate that compared to some state-of-the-art CMOEAs, the proposed algorithm exhibits superior convergence and yields a more uniformly distributed set of feasible Pareto optimal solutions.

The remainder of this article is structured as follows. Section II reviews the classic CMOEAs and decision variable analysis methods, and provides the motivation of this paper. The details of the proposed algorithm are explicated in Section III. Section IV outlines the experimental setup and presents the results. Finally, Section V concludes with a summary of findings and suggestions for future research directions.

II. Related Work

In this section, in addition to introducing the three categories of CMOEAs, some unconstrained multi-objective evolutionary algorithms (MOEAs), which first analyze the correlation between the objectives and the decision variables and then solve the optimization problems based on the correlation [21]–[22], are also introduced. Then, the motivation of the algorithm designed in this paper is stated.

A. The Methods Based on Single Population and Single Stage

In this type of method, the evolutionary process remains undivided into multiple stages, utilizing a single population to progressively approximate the CPF.

NSGA-II-CDP, proposed by Deb *et al.* [23], stands out as a classical method for addressing CMOPs. This approach consistently favors feasible solutions over their infeasible counterparts. However, this inclination often results in the population converging towards local optima, thereby impeding the discovery of Pareto optimal solutions. Considering this limitation, Fan *et al.* [24] introduced angle information into CDP and proposed the MOEA/D-ACDP algorithm, which can utilize infeasible solutions information to increase population diversity and help the population explore in a larger area. In addition, Ma *et al.* [25] employed two distinct methods, CDP and the objective function values of the population, to individually rank the population. Subsequently, they devised an adaptive weighting scheme based on the proportion of feasible solutions in the population to combine the two rankings. This approach aims to leverage both the information from the objective function and maintain the diversity of the population. Similarly, Yu *et al.* [26] designed a dynamic selection preference search algorithm. Notably, the weights assigned to the two rankings are determined using evolutionary algebra. This approach ensures that the objectives receive higher weights in the early stage of evolution, thereby promoting diversity enhancement within the population. Furthermore, to guide the population towards feasible regions from various directions and facilitate gradual convergence to CPF, Ma and Wang [5] introduced a shift-based penalty method named Ship. In this approach, infeasible individuals undergo a shift by considering the distribution of their neighboring feasible individuals.

B. The Methods Based on Two-Stage

This type of method divides the population evolution process into two distinct stages, then each stage has different optimization purposes.

In [27], a push and pull search (PPS) framework was designed. During the push phase, all constraints are disregarded, enabling the exploration of the UPF. Instead, all constraints are considered in the subsequent pull phase, facilitating the population's transition from the UPF back to CPF. Similarly, the first stage of MSCMO [28] is also searching for UPF, but the difference is that constraints are gradually added instead of adding all constraints at once in the second stage to reduce the difficulty of the search. In the study conducted by Tian *et al.* [29], a two-stage evolutionary algorithm was introduced. This algorithm employs distinct evaluation criteria in each stage, and the population exhibits an adaptive capacity to switch between these stages during the evolutionary process, guided by the proportion of feasible solutions. Zhu *et al.* [30] introduced a detect-and-escape mechanism in their study. This mechanism is triggered when the population is identified as being trapped in either an infeasible or local feasible region. Subsequently, the constraint boundary is relaxed to facilitate the population escape from the stagnant state. Yu *et al.* [31] investigated a novel two-stage algorithm. In the initial stage, the optimization objective is to achieve a balance between diversity and convergence. Subsequently, the focus shifts to maintaining the diversity and feasibility of the population in the subsequent stage. Ming *et al.* [32] devised a two-stage framework, where the initial stage is used to search for UPF while maintaining the convergence and diversity of the archive for storing feasible solutions. Subsequently, in the second stage, the purpose is to identify the CPF. It is essential to highlight that the archive derived from the initial stage will be employed as the foundational population for the subsequent phase.

C. The Methods Based on Two-Population

This method utilizes two populations to optimize objectives, and in general, the co-evolutionary approach are employed by these two populations to exchange information and achieve the balance between objectives and constraints.

Tian *et al.* [33] designed a two-population collaborative optimization framework, where one population focuses on both objectives and constraints for searching CPF, while the other population only optimizes the objectives for searching UPF. Li *et al.* [34] introduced a dual-archive optimization algorithm, wherein one convergence archive is employed to ensure the convergence of the population, thereby promoting the exploration of the CPF. Simultaneously, a diversity archive is designed to explore regions that have not been searched by the convergence archive, enhancing the diversity of the population. Liu *et al.* [35] investigated a bidirectional collaborative optimization algorithm, BiCo. In this algorithm, one population employs CDP to explore the CPF from the side of feasible regions, while the other population utilizes the Pareto dominance principle and an angle-based selection mechanism to approach CPF from the side of infeasible regions. In an effort to mitigate resource wastage during the later stages of evolution for the auxiliary population, Liang *et al.* [36] implemented a mechanism wherein the size of the auxiliary population decreases as the evolution progresses. Recently, Qiao *et al.* applied the idea of multitasking to solve CMOPs. In [37], two tasks were formulated. The main task leverages information from feasible solutions to search for the CPF, while the auxiliary task is dedicated to exploring high-quality infeasible solution information. Notably, the constraint boundary of the auxiliary task dynamically changes to maintain a high degree of similarity with the main task. Afterwards, an improved IMTCMO algorithm [38] was developed that utilizes different search methods such as global search and local search to increase population diversity and explore more promising regions.

D. MOEAs Based on Decision Variables Analysis

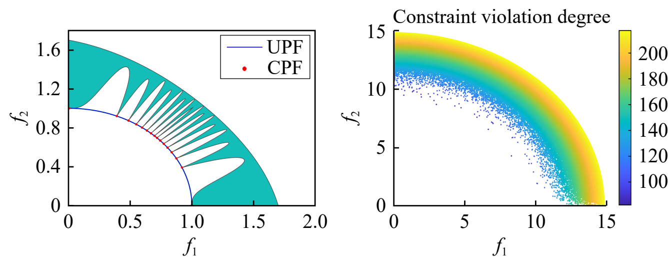
This category of methods is focused on developing mechanisms to analyze the relationship between decision variables and objectives, and subsequently, employing this relationship to address optimization problems.

Ma *et al.* [39] introduced a multi-objective evolutionary algorithm grounded in the analysis of decision variables. This approach entails perturbing decision variables and categorizing them into distance variables, position variables, and mixed variables. Subsequently, distinct operations are applied to different types of variables. Zhang *et al.* [40] developed an evolutionary algorithm by incorporating clustering techniques. In this approach, the decision variables are categorized into convergence-related variables and diversity-related variables. Then specific operations are applied to each type of variable. In the work presented by Ma *et al.* [41], the guidance of reference vectors was integrated into the analysis of decision variables. They devised an adaptive local decision variables analysis method, and subsequently employed an adaptive quantization strategy to optimize the grouped decision variables. By perturbing decision variables, Liu *et al.* [42] classified decision variables into convergence variables and diversity variables based on the consistency of changes in newly generated individuals across different objectives.

E. Motivation

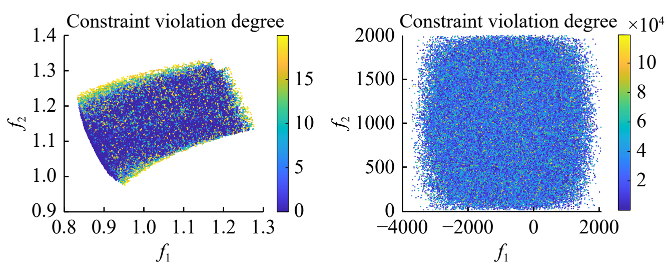
Although a large number of optimization algorithms have been proposed to solve CMOPs, most existing methods focus on solving problems with simple constraints, where the objectives and constraints exhibit a simple positive relationship. As depicted in Fig. 1, the left figure shows the distribution of the feasible regions in the objective space, and the right figure shows the relationship between the objectives and the constraints obtained by uniform sampling of 300 000 times, where the color represents the degrees of constraint violation. It can be observed that there is a positive and straightforward relationship between them. From a global perspective, the changes in constraints are consistent with those of the objectives, as the objectives are optimized, the constraints also gradually decrease. Moreover, the distance between CPF and UPF is relatively close, and the feasible regions only exist in a small area. Based on this observation, existing strategies tend to ignore constraints in the early stage of evolution to quickly approach regions containing CPF. Then, different constraint handling techniques [27], [33] are devised in order to comprehensively investigate the regions lying between CPF and UPF, thereby leading to the identification of the optimal feasible region.

However, for most real-world problems, apart from these simple constraints that have a positive relationship with the objectives, there are also complex constraints that exhibit an intricate relationship with the objectives. In such cases, the problem becomes challenging to solve. As depicted in Fig. 2, two practical problems, the front rail design problem [43] and haverly's pooling problem [44], are uniformly sampled 300 000 times, and the relationships between constraints and objectives in the objective space are given, similarly, the color is employed to reveal the degrees of constraint violation. It can be observed that the relationship between constraints and objectives is difficult to determine and is no longer characterized by a simple positive correlation. Consequently, if the algorithm continues to rely on objective information from UPF, the population is highly likely to become trapped in infeasible or locally feasible regions. Therefore, the algorithms mentioned above do not work well for problems with such complex relationships. In addition, since the essence of optimization is to find the suitable decision variables to satisfy the constraints and optimization objectives, in this paper, we are committed to analyzing the relationship between variables and constraints, and dividing variables into two categories: CR and CI. Afterwards, different optimization strategies are designed to carry out targeted optimization of the two types of variables respectively to achieve different purposes, so as to better search for CPF.



(a) The distribution of feasible regions (b) The relationship between constraint violation and objective function values

Fig. 1. The simple positive relationship between constraints and objectives.



(a) Front rail design (b) Haverly's pooling problem

Fig. 2. The complex relationship between constraints and objectives.

III. Proposed Method

A. DVCEA

The comprehensive framework and procedural details of the proposed algorithm, DVCEA, are illustrated in Fig. 3 and Algorithm 1, respectively. Initially, a population comprising NP individuals is randomly initialized within the decision space. Then, the objective function values and constraint violation degrees of these individuals are computed. Following this, the decision variables undergo classification into two categories using the designed decision variables classification method. Once the decision variables are classified, sequential optimization is carried out on different decision variables in the following loop. First, as shown in Lines 8–13 of Algorithm 1, the constraint-related variables are optimized. In this step, the offspring generation strategy DE/current-to-$p$best/1 is applied to produce superior offspring. Next, an improved ϵ method [37], amalgamating the ϵ constrained method with the

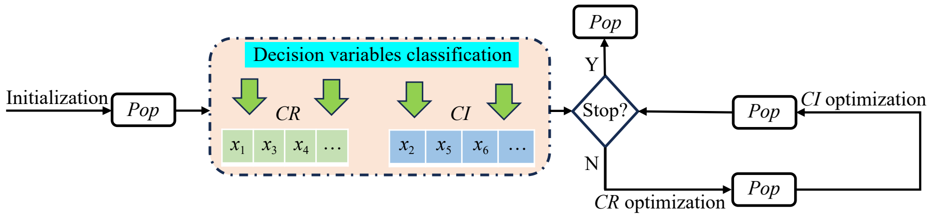


Fig. 3. The framework of DVCEA.

multi-objective method, is applied. This method utilizes multi-objective techniques within the constraint scaling range to augment the population's diversity. It is employed to compute the fitness of each individual within the combined population of parents and offspring. Subsequently, the top NP individuals with superior fitness are retained. Afterwards, the constraint-independent variables are optimized, this is shown in Lines 14–19 of Algorithm 1. The difference is that offspring generation strategy DE/better/1 is first used to generate NP offspring, then the improved ϵ method is employed to calculate the fitness of each individual, and the NP individuals with better fitness are selected in the environmental selection stage. Please note that constraint-independent variables (constraint-related variables) remain unchanged when constraint-related variables (constraint-independent variables) are optimized.

Algorithm 1 The Procedure of DVCEA

Input: NP (population size);
$MaxFes$ (maximum number of function evaluations);
D (problem dimension);
K (the number of selected solutions for decision variables classification);
PE (the number of perturbations on each solution for decision variables classification);
Output: Pop (the final population);

1 $Pop \leftarrow$ Randomly initialize population Pop with NP individuals;
2 Compute the objective function values and constraint violation degrees for the individuals within the population;
3 $Fes = NP$;
4 $Fitness \leftarrow$ Compute the fitness of each individual in Pop using the improved ϵ method;
5 $[CR, CI] \leftarrow$ Decision Variables Classification Method;
6 $Fes = Fes + PE * K * D$;
7 **while** $Fes \le MaxFes$ **do**
// *Constraint-related variables optimization*
8 $Off \leftarrow$ Generate NP offspring by only altering constraint-related variables using the offspring generation strategy DE/current-to-$p$best/1;
9 Evaluate Off;
10 $Pop \leftarrow Pop \cup Off$;
11 $Fitness \leftarrow$ Compute the fitness of each individual in Pop using the improved ϵ method;
12 $Pop \leftarrow$ Select the top NP individuals from Pop based on their fitness values $Fitness$;
13 $Fes = Fes + NP$;
// *Constraint-independent variables optimization*
14 $Off \leftarrow$ Generate NP offspring by only altering constraint-independent variables using the offspring generation strategy DE/better/1;
15 Evaluate Off;
16 $Pop \leftarrow Pop \cup Off$;
17 $Fitness \leftarrow$ Compute the fitness of each individual in Pop using the improved ϵ method;
18 $Pop \leftarrow$ Select the top NP individuals from Pop based on their fitness values $Fitness$;
19 $Fes = Fes + NP$;
20 end while.

In DVCEA, decision variables are first classified into two categories according to their relationship with constraints, i.e., constraint-related variables and constraint-independent variables. Afterwards, these two types of decision variables undergo separate optimization processes, and the advantages are twofold. The first advantage is the reduction of search difficulties caused by the increase in the dimension of the problem. If all variables are optimized together, the search space will grow exponentially due to the increase in dimensions, and the optimization is no longer targeted if variables from different categories are optimized together, and variables may mislead each other, making it challenging for the population to find the correct search direction. While when the decision variables are optimized separately, which is equivalent to the dimension of the problem is reduced, which makes it easier for the algorithm to search the decision space and find promising regions. In addition, separate optimizations allow for more refined operations for different variables. The second advantage is that when the decision variables are separated, different specific optimization operations can be carried out on different decision variables. In this way, the search of the algorithm is no longer blind, which helps the algorithm to search more efficiently.

Specifically, when optimizing these two types of decision variables, the different offspring generation strategies is used to generate promising offsprings. DE/current-to-$p$best/1 is considered for the former, the reason is that these variables have a significant impact on the degree of individual constraint violation, and it is expected to obtain individuals with smaller constraint violation degrees and accelerate the convergence of the population. Indisputably, DE/current-to-$p$best/1 has the ability to accelerate population convergence. For the latter, its impact on the degrees of constraint violation of individuals is relatively small, so it is desirable that these variables can be optimized to help the population achieve excellent diversity. Based on this, DE/better/1 is employed to generate offspring of constraint-independent variables. In addition, the same fitness value calculation method, improved ϵ method, is employed. The reason is that it can increase population diversity in the early evolution stage, and facilitates population feasibility in the later evolution stage. The specific

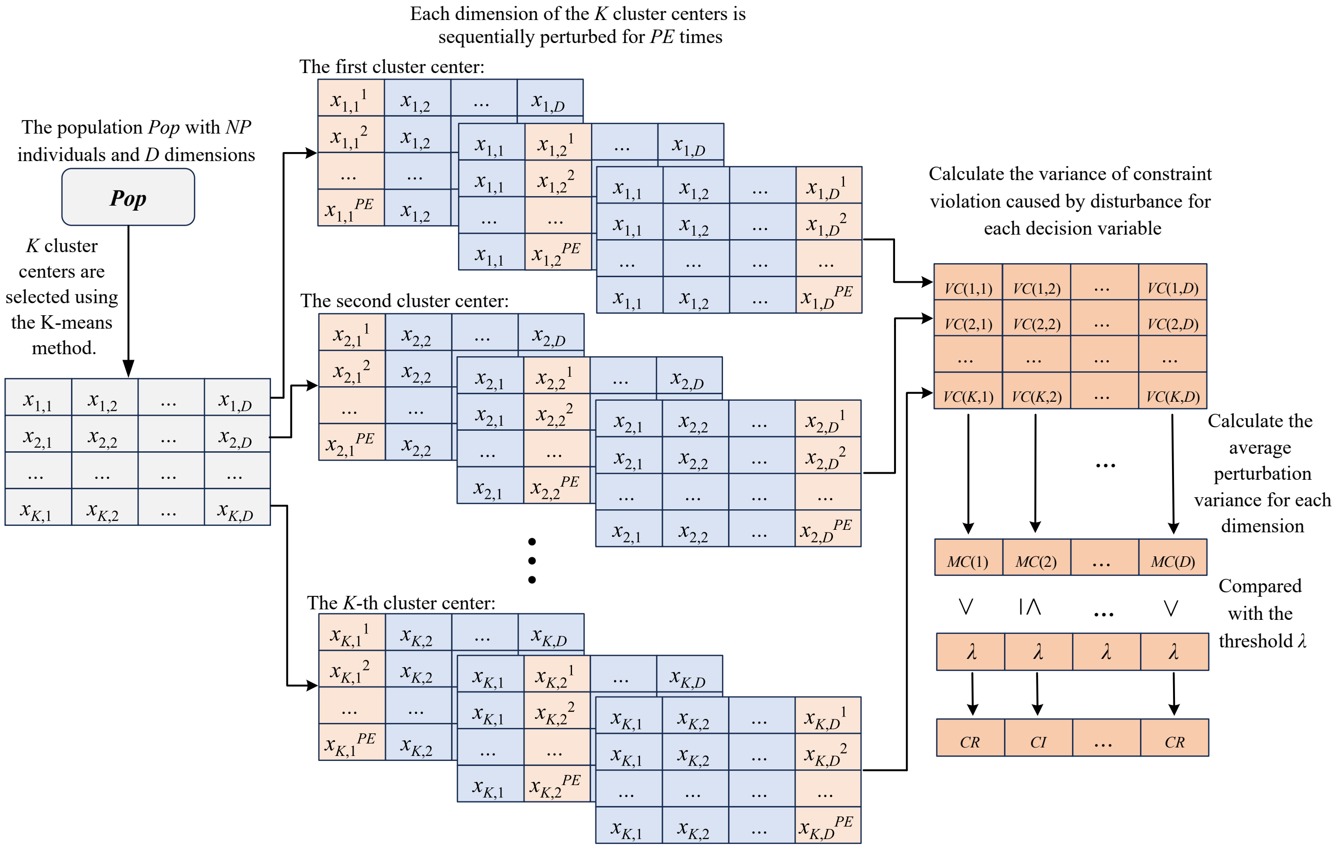


Fig. 4. The specific steps of the decision variables classification method.

operations are introduced in the Section III-C.

B. Decision Variables Classification Method

In the proposed algorithm, a preparatory work is needed before optimizing the variables, that is, classifying the decision variables according to their characteristics, so that a specific optimization can be performed after the classification. Therefore, how to make a reasonable classification of variables is extremely important.

Algorithm 2 and Fig. 4 show the specific steps of the decision variables classification method. For any problem, the following steps are taken to classify the decision variables.

Algorithm 2 Decision Variables Classification Method

Input: *Pop* (the population);
NP (population size);
D (problem dimension);
K (number of selected solutions for decision variables classification);
PE (number of perturbations on each solution for decision variables classification);
λ (decision variables category classification threshold);
Output: CR (constraint-related variables);
CI (constraint-independent variables);

$C \leftarrow$ K-means (Pop, K);
for $k = 1 : K$ **do**
  for $d = 1 : D$ **do**
    Apply uniform perturbations PE times to the d-th decision variable of the k-th individual in C;
    $VC(k,d) \leftarrow$ Compute the variance of constraint violation caused by disturbance for each decision variable;
  end for
end for
$MC \leftarrow$ Calculate the average perturbation variance for each dimension;
$CR \leftarrow$ Identify the decision variables that meet the condition $MC > \lambda$;
$CI \leftarrow$ Identify the decision variables that meet the condition $MC \leq \lambda$.

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1) Selection of Disturbed Individuals: At this step, K individuals are selected and their decision variables are disturbed. In order to select diverse individuals, a K-means clustering method is used. In this way, K clustering centers with preferable dispersion are obtained, and then the K individuals are perturbed to prevent the situation where the perturbed individuals are close to each other and the degree of change remains similar after perturbation. The first column of Fig. 4 shows K cluster centers after clustering.

2) Perturbation of Decision Variables: PE uniform perturbations are applied to each decision variable of these individuals to observe the changes in constraints, please note that when one dimension of an individual is perturbed, the other dimensions remain unchanged. The second column of Fig. 4 illustrates the perturbation of decision variables for each individual, where the light pink color represents the perturbed decision variable. Specifically, for each decision variable, the perturbations are as follows:

$$x_d^{pe+1} = x_d^l + pe \times \frac{x_d^u - x_d^l}{PE - 1}, \quad pe = 0, \ldots, PE - 1 \tag{4}$$

where x_d^{pe+1} is the value of the decision variable after the $(pe+1)$-th perturbation. x_d^l and x_d^u represent the lower and upper bounds of decision variables of the dth dimension, respectively.

3) Classification of Decision Variables: After perturbing each decision variable for each individual, the average of the variance of the constraint violation degree is calculated and compared to the threshold λ to classify the decision variables into different categories. The third column of Fig. 4 shows specific details, firstly, the variance of constraint violation degree caused by disturbance for each decision variable is calculated and recorded in VC, where VC is a matrix of $K \times D$. After all decision variables are perturbed and the variance of constraint violation degree is recorded, the average values of the variances are calculated and stored in MC, where MC is a matrix of $1 \times D$. If a decision variable has a significant impact on the constraint, then the average value of its perturbation variance should also be large, and vice versa, its average value should be small. Therefore, a decision variables classification threshold λ is set, which is a small positive number and is set to 0.00001 in this study. A decision variable is categorized as a constraint-related variable if $MC > \lambda$, otherwise, it is considered a constraint-independent variable. In addition, a specific example of applying decision variable analysis method to a practical problem is provided in Section II of the supplementary file (supplementary file can be available at: https://www.ieee-jas.net/article/doi/10.1109/JAS.2025.125276?pageType=en).

C. Decision Variables Optimization Strategies

The optimization of decision variables is a crucial component of population evolution, and it is the driving force to push the population towards CPF. In DVCEA, two types of variables, CR and CI, are specified. CR has a significant impact on constraints, so it is hoped that such variables can be optimized to find feasible regions and achieve rapid convergence of the population. CI has a very small impact on constraints, so it can be optimized to increase the diversity of the population. Based on this, in this paper, two variables optimization strategies with different properties are considered to optimize different types of variables, respectively. Their processes are illustrated in Algorithms 3 and 4, respectively.

For constraint-related variables, the fitness of each individual is first computed based on the improved ϵ method. After that the former $NP * p$ individuals with the smallest fitness values are designated as $pbest$. Then, the DE/current-to-$p$best/1 strategy is employed by each individual to generate offspring:

1) DE/current-to-$p$best/1:

$$\vec{v}_i = \vec{x}_i + F * (\vec{x}_{pb} - \vec{x}_i) + F * (\vec{x}_1 - \vec{x}_2) \tag{5}$$

where $\vec{v}_i$ denotes the offspring individual produced by the individual $\vec{x}_i$, $\vec{x}_1$ and $\vec{x}_2$ denote two different individuals randomly chosen from the population. $\vec{x}_{pb}$ is an individual randomly selected from the best individuals $pbest$. F is a scaling factor whose value is randomly taken from the scaling pool {0.6, 0.8, 1.0}. Fig. 5(a) shows the schematic diagram of the DE/current-to-$p$best/1 strategy, which can assist individuals in the population to learn from superior individuals, guide the population to evolve towards promising directions, and accelerate the population convergence.

Algorithm 3 Optimization Strategy for CR

Input: Pop (the population);
NP (population size);
CR (constraint-related variables);
p (a parameter);
$Fitness$ (the fitness of each individual);
F (scale factor);
Output: Off (offspring population);
$pbest \leftarrow$ Select the top $NP \times p$ individuals with the smallest fitness values;
for $i = 1 : NP$ **do**
  $\{\vec{x}_1, \vec{x}_2\} \leftarrow$ Randomly select two distinct individuals from the population Pop;
  $pb \leftarrow$ Randomly select an individual from the $pbest$;
  $\vec{v}_i = \vec{x}_i + F * (\vec{x}_{pb} - \vec{x}_i) + F * (\vec{x}_1 - \vec{x}_2)$;
  $v\vec{v}_i = \vec{x}_i$;
  $v\vec{v}_i(CR) = \vec{v}_i(CR)$;
  $Off = Off \bigcup \{v\vec{v}_i\}$;
end for.

Algorithm 4 Optimization Strategy for CI

Input: Pop (the population);
D (problem dimension);
CI (constraint-independent variables);
$Fitness$ (the fitness of each individual);
F (scale factor);
Output: Off (offspring population);
while $|Pop| > 1$ **do**
  $\{\vec{x}_y, \vec{x}_z\} \leftarrow$ Randomly select two individuals x_y and x_z from the population Pop;
  if $Fitness(x_y) < Fitness(x_z)$ **then**
    $\vec{x}_b = \vec{x}_y$;
    $\vec{x}_w = x_z$;
  else
    $\vec{x}_b = \vec{x}_z$;
    $\vec{x}_w = \vec{x}_y$;
  end if
  $\vec{v}_b = \vec{x}_b + F * (\vec{x}_b - \vec{x}_w)$;
  $index = \text{find}(rand(1, D) < 0.5)$;
  $\vec{v}_w = (\vec{x}_w(index) = \vec{x}_b(index))$;
  $v\vec{v}_b = \vec{x}_b$;
  $v\vec{v}_b(CI) = \vec{v}_b(CI)$;
  $v\vec{v}_w = \vec{x}_w$;
  $v\vec{v}_w(CI) = \vec{v}_w(CI)$;
  $Pop = Pop \backslash \{\vec{x}_y, \vec{x}_z\}$;
  $Off = Off \bigcup \{v\vec{v}_b, v\vec{v}_w\}$;
end while.

For each individual $\vec{x}_i$, after generating the vector $\vec{v}_i$ by using the DE/current-to-$p$best/1 strategy, the offspring individual $v\vec{v}_i$ is obtained by replacing the constraint-related variables in the parent vector $\vec{x}_i$ with those in vector $\vec{v}_i$. That is to say, when optimizing CR, only CR changes, while CI remains unchanged.

Since CR is more sensitive to constraint values, so it is hoped that these variables can be optimized to facilitate rapid convergence of the population and enter the feasible regions. Therefore, the convergence strategy DE/current-to-$p$best/1 is employed. Furthermore, the improved ϵ method is used to enhance the diversity of the population, prevent the population from getting trapped in a local feasible region in the early stage, and promote population into feasible areas as the evolution progresses.

For constraint-independent variables, the fitness value of each individual in the population is initially calculated using the improved ϵ method. After that, the loop continues until the size of the population Pop is 0. At the start, two individuals $\vec{x}_y$ and $\vec{x}_z$ are randomly selected from the Pop and their fitness values are compared, the better individual is considered as $\vec{x}_b$ and the worse as $\vec{x}_w$, for the better individual, the following operation are performed:

2) DE/better/1:

$$\vec{v}_b = \vec{x}_b + F * (\vec{x}_b - \vec{x}_w) \tag{6}$$

where $\vec{v}_b$ denotes the offspring individual produced by the better individual $\vec{x}_b$. In contrast, for the worse individuals, each of their dimensions has a probability of 0.5 to receive information from better individuals. As shown in lines 12 and 13 of Algorithm 4, a random number is first generated for each dimension, and if the random number is less than 0.5, the information of this dimension from the better individual is directly given to the worse individual. Fig. 5(b) illustrates the schematic diagram of the DE/better/1 strategy, since paired individuals are randomly matched, the learning directions of the population can be increased, thereby improving the diversity of the population and enabling the population to explore more promising areas.

For every two parent individuals $\vec{x}_y$ and $\vec{x}_z$, vectors $\vec{v}_b$ and $\vec{v}_w$ are obtained by using (6) and dimension assignment method, respectively. Afterwards, the offspring individuals $v\vec{v}_b$ and $v\vec{v}_w$ are generated by replacing the constraint-independent variables in $\vec{x}_y$ and $\vec{x}_z$ with those in the vectors $\vec{v}_b$ and $\vec{v}_w$, respectively. Moreover, individuals $\vec{x}_y$ and $\vec{x}_z$ will be removed from the population. Similarly, only CI is changed and CR remains the same when optimizing CI.

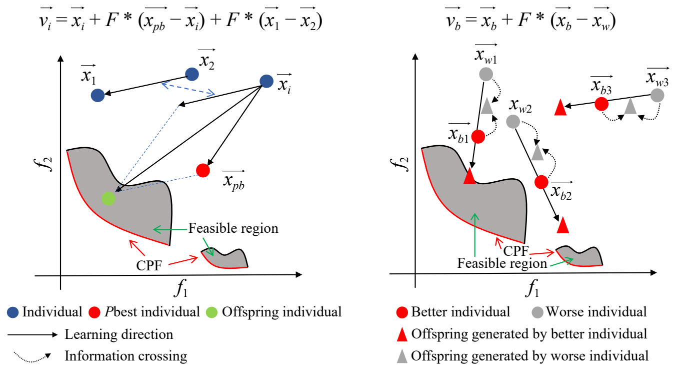
CI is insensitive to constraints, so it is hoped that these variables can be optimized to increase the diversity of the population and facilitate exploration across a broader range of areas. In the proposed offspring generation strategy, the learning directions are diverse, which contributes to the diversity of the population. Moreover, because the worse individuals learn from the better ones, it is beneficial to the convergence of the population to a certain extent. For the better individual, it will move further in the direction of promising evolution, which plays a significant role in finding promising regions. For the worse individual, it absorbs valuable information from the better individual, in this way, the regions ignored by the better individual can be explored and the diversity of the population is improved.

D. Algorithm Analysis

For a given problem, DVCEA first classifies the decision variables of the problem into constraint-related variables and constraint-independent variables, subsequently, distinct optimization strategies are applied to each category. Indisputably, whether different classification results will affect the performance of the algorithm should be investigated, which is also the prerequisite for the algorithm to classify and optimize decision variables. Herein, several problems, SDC5, SDC7, SDC8, SDC11, and SDC12 [38], are selected to investigate the impact of classification accuracy on algorithm performance. In addition, if the ratio between the count of constraint-related variables and the count of constraint-independent variables is changed, the classification of variables should also be adjusted accordingly, and the results presented by the algorithm are also different. At this point, the classification method used by DVCEA faces the challenge. Therefore, to assess the stability of the devised classification method and the effect of problems with different proportions on algorithm performance, the same problems mentioned above are employed to adjust the proportion between the count of constraint-related variables and the count of constraint-independent variables to generate new variant problems. Please note that the total number of variables remains unchanged, only the number of constraint-related and constraint-independent variables is changed, and the experimental settings are given in Section VI-A. Furthermore, the influence of different number of decision variables on the performance of the algorithm is also investigated, and the experimental results and analysis are displayed in the supplementary file.

1) The Effect of Classification Accuracy on Algorithm Performance: In order to investigate the impact of varying classification accuracy on algorithm performance, several different classifications are set in advance, that is, the decision variables classification method is no longer used, and different classifications are set artificially so that their accuracies are 0.2, 0.4, 0.6, 0.8, 1.0, respectively. The changes in algorithm performance with different classification accuracies are presented in Fig. 6, and the specific average IGD^+ results are provided in Table S-1 in the supplementary file. As a whole, it can be observed that with the improvement of classification accuracy, the average IGD^+ values show a decreasing trend, indicating that the performance of the algorithm will increase with the improvement of classification accuracy. Therefore, for the problems that need to be solved, higher classification accuracy is expected when classifying decision variables.

2) Classification Accuracy Analysis of DVCEA: The classification accuracy of the classification method used in DVCEA on several new variant problems is shown in Fig. S-1 in the supplementary file, where SDC_0.3, SDC_0.5, and SDC_0.8 respectively indicate that the proportion of constraint-related variables to all variables is 0.3, 0.5, and 0.8. It can be seen from the results that regardless of how the proportion of different variables in the problem is adjusted, DVCEA can achieve 100% classification accuracy on all variant problems. This is mainly because DVCEA perturbs each variable individually, so the change values of each variable can be recorded. Consequently, altering the proportions between different variables does not impact the classification accuracy. In



(a) DE/current-to-pbest/1 strategy diagram (b) DE/better/1 strategy diagram

Fig. 5. Offspring generation strategies diagram.

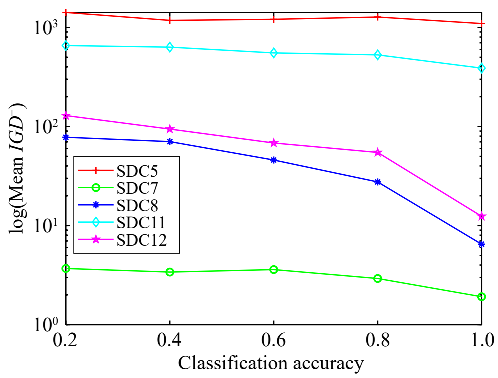


Fig. 6. Diagram of the effect of different classification accuracies on algorithm performance.

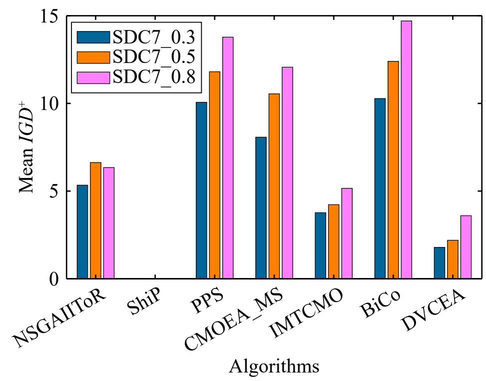


Fig. 7. The average IGD^+ diagram of the algorithms on the SDC7 variant problem.

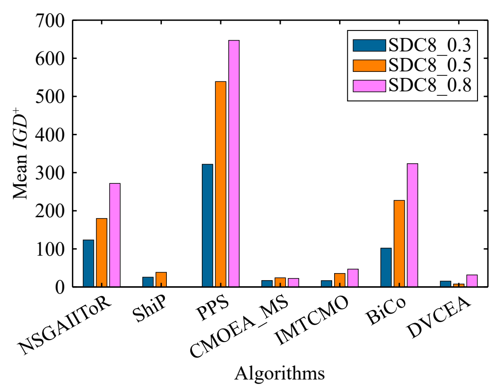


Fig. 8. The average IGD^+ diagram of the algorithms on the SDC8 variant problem.

a word, this also proves the stability of the classification method, which can better prepare for the next optimization.

3) Performance Analysis of DVCEA: The mean IGD^+ results HV results are shown in Tables S-2 and S-3 respectively in the supplementary file. According to the experimental results, DVCEA significantly outperforms the comparison algorithm on 15, 14, 15, 12, 9, and 15 variant problems, respectively. In addition, the performance of the algorithm decreases with the increase of the ratio between the constraint-related variables and the constraint-independent variables. In fact, as the number of constraint-related variables increases, the constraints of the problem will become complicated and more difficult to satisfy, resulting in an increase in the difficulty of the problem. Furthermore, to provide a clearer observation of the algorithmic performance changes, the average IGD^+ graphs of SDC7 and SDC8 variant problems are drawn in Figs. 7 and 8, while IGD^+ graphs for the remaining problems are provided in the supplementary file. It is noteworthy that ShiP does not identify feasible solutions, so its value is not plotted. As can be seen from the figures, the IGD^+ values increases with the difficulty of the problems. However, on the whole, DVCEA achieves the superior performance on all problems of different difficulty, and its performance stability is relatively advantageous.

E. Computational Complexity

In the proposed algorithm, the most important operations include decision variables classification, offspring generation strategies, and environment selection. Specifically, the computational complexity of decision variables classification is $O(K \cdot D)$, the computational complexity of offspring generation strategy is $O(NP \cdot D)$, and the maximum computational complexity of environment selection is $O(M \cdot NP^2)$. Therefore, for DVCEA, its total computational complexity is $O(K \cdot D) + 2 \cdot (O(NP \cdot D)) + O(M \cdot NP^2)) = O(M \cdot NP^2)$, which aligns with the computational complexity of most CMOEAs.

IV. Experimental Studies

This section conducts numerous experiments to assess the performance of the proposed algorithm and validate the effectiveness of the designed strategies. Firstly, the setup of the experiments is introduced, including the comparison algorithms, the test problems used, and some parameter settings. After that, the comparison experiments with peer algorithms are carried out. Subsequently, various variant algorithms are established to assess the effectiveness of the designed strategies. The experimental setup includes tests on 20 real-world problems to evaluate the algorithm's capability to address practical concerns.

A. Experimental Setup

Six advanced CMOEAs are selected for comparison experiment with DVCEA, which are NSGAIIToR [25], ShiP [5], PPS [27], CMOEA_MS [29], IMTCMO [38], and BiCo [35], and they belong to different categories. NSGAIIToR and ShiP are the methods based on single population and single stage, PPS and CMOEA_MS belong to the methods based on two-stage, and the last two algorithms are the methods based on two-population.

The SDC test set [38], ZXH_CF test set [45], and 20 real-world problems [20] are selected for experimental comparison. The SDC test set contains 15 problems with complex constraints, which exhibit a chaotic relationship between the objectives and constraints as shown in Fig. 2, they are selected to test the ability of the designed algorithm to solve complex constraints problems. At the same time, to test the performance of the proposed algorithm on simple constraints problems, where the trend of changes in objectives and constraints shows a simple positive relationship as shown in Fig. 1, the ZXH_CF test set containing 16 problems is selected. In addition, 20 practical problems are selected to observe the ability of the designed algorithm to solve real-world problems, which include both complex and simple constraints, to comprehensively test the performance of the designed algorithm.

For a fair comparison, all algorithms undergo 25 runs on each test problem, the maximum number of evaluations $MaxFes$ is set to 300 000, population size NP is 100, and the dimension of each problem is set to 100. For six comparison algorithms, the remaining parameters are maintained in accordance with the original literature specifications. For DVCEA, K (the number of selected solutions for decision variables classification) is set to 5, and PE (the number of perturbations for each dimension) is configured at 4, and their effectiveness is demonstrated in the following experiments. The value of parameter p in DE/current-to-$p$best/1 strategy is 0.1 [38], the scaling factor F in both optimization strategies is randomly selected from the scaling pool {0.6, 0.8, 1.0} in each generation [46], and the threshold λ is set to 0.00001. In addition, all experiments are performed on PlatEMO 4.2 version [47].

In addition, three performance indicators, inverted generational distance based on modified distance calculation (IGD^+) [48], Hypervolume (HV) [49], and feasible solutions rate (FSR), are employed to test the performance of various CMOEAs. IGD^+ can calculate the distance from the real reference points to the final population obtained by the algorithm, offering insights into the convergence and diversity of the population, smaller IGD^+ values correspond to better algorithm performance. HV can calculate the volume enclosed by the final population discovered by the algorithm and a reference point, providing additional indicators of convergence and diversity. Higher HV values signify superior algorithm performance. Please note that the objectives of the feasible solutions are normalized to the range [0, 1], and the reference points are set to [1.1, 1.1] when the HV is computed. FSR calculates the proportion of average feasible solutions in the final population over 25 runs, with a larger value indicating better algorithm performance. Additionally, the Wilcoxon rank sum test with a significance level of 0.05 is used to evaluate whether there exists a significant difference between algorithms, in the result tables, the symbol "+", "–", and "=" respectively indicate that the compared algorithm is significantly better than, significantly worse than, and similar to the proposed algorithm, and the shadow indicates that the algorithm achieves the best average value on the corresponding problem.

B. Comparison Experiments With Peer Algorithms

Table I presents the specific IGD^+ results of DVCEA compared with other algorithms. The optimal result for each problem is highlighted in gray. If an algorithm fails to achieve a 100% feasibility rate on a problem, the result is presented in the form of "NaN (FSR)". From Table I, DVCEA significantly outperforms NSGAIIToR, ShiP, PPS, CMOEA_MS, IMTCMO, and BiCo on 24, 26, 30, 23, 24, and 19 problems, respectively, whereas the comparison algorithms outperform DVCEA significantly on 1, 1, 0, 2, 1, and 2 problems, respectively.

NSGAIIToR ranks each individual in the population based on the Pareto dominance principle and CDP, respectively, and then an adaptive weight is used to weight the two rankings, leading the population to focus on exploring objectives in the early stage and giving priority to constraint satisfaction in the later stage. In this way, excellent performance can be achieved in those problems where the constraint fitness landscape is not complicated. However, due to the consistently higher weight assigned to constraints, it cannot explore broader regions within the search space, making it less effective on more complex problems, so it can not get better solutions for problems with complex constraints.

For the ShiP, infeasible solutions are penalized and transferred, with the extent of the transfer determined by the distribution and proportion of feasible solutions in the current population. In this way, infeasible solutions can be transferred to promising feasible regions. However, there is also a certain risk associated with situations where an infeasible solution, anticipated to enter the feasible regions, is situated in a position dominated by the local optimal feasible region. In such cases, the penalty mechanism may lead to the solution being transferred to other regions, significantly compromising the algorithm's performance. Additionally, due to these improper transfers, the population may struggle to enter feasible regions, making it difficult to search for feasible solutions. Therefore, ShiP cannot achieve the feasibility rate of 100% on most complex problems.

PPS is a two-stage algorithm where the primary objective in

TABLE I
MEAN IGD^{+} RESULTS OF DVCEA AND COMPARED ALGORITHMS ON SDC AND ZXH_CF PROBLEMS

	The methods based on single population and single stage		The methods based on two-stage		The methods based on two-population		The proposed algorithm
Problem	NSGAIIToR	ShiP	PPS	CMOEA_MS	IMTCMO	BiCo	DVCEA
SDC1	1.0482e+1 (1.22e+0) =	1.0405e+1 (9.53e−1) =	1.3677e+1 (2.25e+0) −	1.0080e+1 (8.35e−1) =	1.1445e+1 (1.69e+0) −	1.1022e+1 (1.82e+0) =	1.0148e+1 (1.41e−1)
SDC2	3.2488e−1 (7.15e−2) =	3.4548e−1 (9.21e−2) =	1.1148e+0 (8.02e−2) −	5.5370e−1 (8.13e−2) −	5.5840e−1 (5.81e−2) −	2.7082e−1 (1.52e−1) =	3.8472e−1 (1.43e−1)
SDC3	1.3523e+0 (7.48e−2) =	NaN (0%) −	1.9310e+0 (1.60e−1) −	2.1957e+0 (8.48e−2) −	1.5018e+0 (1.68e−1) −	2.0061e+0 (2.61e−1) −	1.3433e+0 (6.78e−2)
SDC4	NaN (80%) −	NaN (0%) −	1.9996e+0 (1.52e−1) −	2.6723e+2 (1.11e+2) −	1.4805e+0 (1.17e−1) =	1.8436e+2 (9.93e+1) −	1.4434e+0 (5.69e−2)
SDC5	1.7197e+3 (2.32e+2) −	1.7046e+3 (1.32e+2) −	2.0192e+3 (3.90e+2) −	1.3921e+3 (1.79e+2) −	1.7398e+3 (2.75e+2) −	1.7076e+3 (2.39e+2) −	1.1389e+3 (2.05e+2)
SDC6	NaN (0%) −	NaN (0.7%) −	4.0847e+2 (2.55e+2) =	1.8905e+3 (1.09e+3) −	8.9516e+1 (1.50e+2) +	7.0441e+2 (4.14e+2) =	3.4798e+2 (3.57e+2)
SDC7	6.6283e+0 (7.73e−1) −	NaN (0%) −	1.1807e+1 (5.45e−1) −	1.0548e+1 (1.93e+0) −	4.2258e+0 (1.01e+0) −	1.2400e+1 (5.37e−1) −	2.1953e+0 (2.83e−1)
SDC8	1.7979e+2 (7.97e+1) −	3.8513e+1 (1.37e+1) −	5.3891e+2 (9.83e+1) −	2.4159e+1 (1.23e+1) −	3.5584e+1 (2.45e+1) −	2.2718e+2 (5.72e+1) −	7.8429e+0 (8.68e+0)
SDC9	7.5213e+2 (1.28e+2) −	NaN (0%) −	2.1252e+3 (3.70e+2) −	NaN (80%) −	2.5914e+3 (2.58e+2) −	1.9100e+3 (1.75e+2) −	4.5855e+2 (9.38e+1)
SDC10	9.9137e+0 (4.04e−1) =	1.0060e+1 (9.33e−1) =	1.0265e+1 (4.42e−1) −	1.0278e+1 (1.16e+0) =	1.0419e+1 (1.05e+0) =	1.0130e+1 (7.67e−1) =	9.9617e+0 (2.66e−1)
SDC11	7.8762e+2 (1.48e+2) −	7.0951e+2 (1.45e+2) −	8.2519e+2 (1.39e+2) −	6.3029e+2 (1.33e+2) −	3.8264e+2 (5.12e+1) =	6.0465e+2 (1.18e+2) −	4.0670e+2 (8.43e+1)
SDC12	1.7621e+2 (4.80e+1) −	1.1114e+2 (4.23e+1) −	7.7425e+2 (8.98e+1) −	1.0131e+2 (3.99e+1) −	5.6977e+1 (2.76e+1) −	1.2497e+2 (1.92e+1) −	8.3027e+0 (7.46e+0)
SDC13	1.2787e+1 (2.00e+0) =	NaN (0%) −	1.4451e+1 (5.73e−1) −	NaN (60%) −	1.6156e+1 (2.66e+0) −	1.6297e+1 (3.37e+0) −	1.2645e+1 (7.26e−1)
SDC14	NaN (0%) −	NaN (0%) −	NaN (60%) −	NaN (0%) −	NaN (80%) =	NaN (20%) −	NaN (80%)
SDC15	9.8896e+0 (1.76e−1) +	9.7641e+0 (2.06e−1) +	2.2576e+1 (1.58e+0) −	9.8215e+0 (2.25e−1) +	1.1637e+1 (9.39e−1) −	9.9328e+0 (1.71e−1) +	1.0083e+1 (4.64e−2)
ZXH_CF1	5.9739e−2 (3.57e−3) −	6.1957e−2 (4.03e−3) −	1.2767e−1 (1.03e−2) −	3.9868e−2 (8.22e−4) =	7.8827e−2 (3.03e−3) −	4.0442e−2 (8.82e−4) =	4.0332e−2 (1.21e−3)
ZXH_CF2	4.6391e−1 (7.40e−1) =	4.6533e−1 (6.52e−1) =	8.8961e−1 (5.93e−1) −	5.1912e−1 (6.94e−1) =	6.7080e−1 (4.80e−1) −	4.5271e−1 (6.61e−1) =	1.6561e−1 (2.76e−1)
ZXH_CF3	6.9304e−2 (9.73e−3) −	1.5229e−1 (2.56e−1) −	NaN (80%) −	5.2402e−2 (2.26e−3) +	1.3805e−1 (7.00e−3) −	4.9694e−2 (3.56e−3) +	5.8004e−2 (2.98e−3)
ZXH_CF4	7.5354e−2 (7.40e−2) −	1.8879e−1 (2.45e−1) −	1.0935e−1 (4.80e−2) −	7.7504e−2 (7.02e−2) −	1.1656e−1 (6.03e−2) −	5.3792e−2 (3.49e−2) −	3.3352e−2 (1.70e−3)
ZXH_CF5	6.6818e−1 (5.08e−1) −	3.2042e−1 (4.10e−1) =	5.6343e−1 (3.51e−1) −	NaN (65%) −	4.2512e−1 (2.90e−1) −	7.8578e−1 (5.13e−1) −	7.1258e−2 (9.07e−2)
ZXH_CF6	3.5219e−2 (3.51e−3) −	3.7091e−2 (8.61e−4) −	4.9463e−2 (2.14e−3) −	2.2899e−2 (1.44e−3) =	4.9045e−2 (1.76e−3) −	2.3380e−2 (1.09e−3) =	2.3469e−2 (5.39e−4)
ZXH_CF7	1.1793e−1 (2.59e−1) −	3.0285e−1 (3.96e−1) −	1.0477e−1 (4.53e−2) −	NaN (34.7%) −	5.3105e−2 (1.75e−3) −	4.5899e−2 (4.03e−2) −	2.1831e−2 (7.05e−4)
ZXH_CF8	4.2447e−2 (1.81e−3) −	4.3745e−2 (2.64e−3) −	1.3528e−1 (6.15e−2) −	1.8647e−1 (2.55e−2) −	8.8635e−2 (2.43e−3) −	3.5083e−2 (1.12e−3) =	3.5995e−2 (1.31e−3)
ZXH_CF9	1.2233e−1 (1.56e−1) −	2.4941e−2 (1.87e−3) −	7.8452e−2 (5.84e−2) −	1.6981e−2 (4.06e−3) −	4.8729e−2 (2.45e−3) −	1.6069e−2 (8.38e−4) −	1.4022e−2 (5.62e−4)
ZXH_CF10	2.1531e−1 (2.12e−1) −	3.2950e−2 (3.75e−2) −	3.5174e−1 (1.37e−1) −	2.3871e−1 (1.35e−1) −	5.8384e−2 (5.52e−2) −	3.2476e−2 (5.32e−2) −	1.2401e−2 (2.86e−4)
ZXH_CF11	3.1942e−1 (2.45e−1) −	1.6997e−2 (1.10e−3) −	4.2931e−1 (1.38e−1) −	9.7453e−3 (3.28e−4) −	2.2268e−2 (9.75e−4) −	1.0855e−2 (4.48e−4) −	9.0386e−3 (2.57e−4)
ZXH_CF12	5.1737e−1 (2.15e−1) −	3.3519e−1 (3.57e−1) =	5.7525e−1 (2.91e−1) −	6.3331e−1 (5.01e−1) −	3.4388e−1 (3.25e−1) =	3.8298e−1 (4.68e−1) =	1.6745e−1 (2.01e−1)
ZXH_CF13	2.2162e−1 (2.15e−1) −	2.5925e−2 (5.58e−2) −	2.2429e−1 (2.18e−1) −	1.5456e−1 (1.41e−1) −	3.3733e−3 (1.92e−4) −	1.9620e−2 (5.46e−2) −	1.1719e−3 (5.62e−5)
ZXH_CF14	9.4005e−3 (1.82e−2) −	3.7953e−3 (1.93e−4) −	1.2553e−2 (9.57e−4) −	2.0339e−2 (2.78e−2) −	8.7860e−3 (3.87e−4) −	3.4172e−3 (2.17e−4) −	2.1886e−3 (5.52e−5)
ZXH_CF15	3.3008e−1 (2.43e−1) −	2.5912e−1 (2.82e−1) −	3.7649e−1 (2.16e−1) −	2.3255e−1 (4.90e−1) =	1.8003e−1 (2.59e−1) =	9.1112e−2 (1.23e−1) =	6.3026e−2 (1.07e−1)
ZXH_CF16	2.2003e−2 (2.14e−2) −	1.7875e−2 (2.10e−2) −	2.5459e−2 (1.57e−2) −	3.9151e−2 (1.23e−2) −	2.4877e−3 (9.13e−5) −	1.2233e−3 (4.63e−5) −	1.1188e−3 (4.88e−5)
+/−/=	1/24/6	1/26/4	0/30/1	2/23/6	1/24/6	2/19/10	

the first stage is to search for the UPF. Subsequently, in its second stage, a modified ϵ method is employed to encourage the population to search from the UPF towards the CPF. Because all constraints are ignored in the first phase, the diversity of the population is enhanced. The full utilization of UPF information helps the population explore higher-quality regions. As a result, it demonstrates satisfactory performance on those problems where UPF and CPF are in the same search direction. However, in these more complex problems, the relationship between the two is not obvious, and the UPF information plays a small role in the search of CPF, so the results obtained by PPS are unsatisfactory.

CMOEA_MS strategically divides the algorithm's evolution process into two stages, guided by the proportion of feasible solutions within the population. When the proportion falls below a predefined threshold, equal weight is assigned to objectives and constraints. Otherwise, constraints are given higher priority. This adaptive strategy facilitates the population's traversal through infeasible regions toward the vicinity of the optimal feasible domain. Nevertheless, as the weight assigned to constraints remains equal to or greater than that of objectives throughout both stages, there may be limitations in promoting population diversity, which will hinder the population's search to some extent. As a result, it fails to achieve the desired effect in problems with complex constraint landscapes and very large infeasible regions.

IMTCMO is an algorithm based on two-population, in which the primary population uses CDP to ensure the feasibility and the secondary population employs an improved ϵ method to enhance population diversity. In addition, both populations perform local searches to further increase population diversity and global searches to improve population convergence. The combination of multiple search strategies enables the population to explore a broader range of regions, leading to excellent results in some test problems. However, this also leads to more complex algorithm operations, moreover, only half of the individual information of IMTCMO is utilized during each evolution, which leads to the neglect of some valuable knowledge. As a consequence, the performance of IMTCMO is not desired in the test problems with very complex nonlinear relationships between decision variables.

For another algorithm, BiCo, which utilizes two populations, the primary population employs CDP to approach the CPF from one side of the feasible regions. Simultaneously, its archive population regards constraint violation as a new objective, with the aim of exploring infeasible solutions characterized by smaller constraint violations, thus approaching the CPF from one side of the infeasible regions. Additionally, it utilizes angle-based environmental selection to enhance population diversity, making it superior to DVCEA in terms of feasibility rates. However, once the archive population falls into the locally feasible region, it is challenging to escape this situation due to the constraint value is difficult to dominate, resulting in unsatisfactory performance on CMOPs with complex constraints.

For DVCEA, the decision variables with different characteristics are first optimized separately, which reduces the difficulty of optimization caused by the increase of dimensions and makes the optimization more targeted. Firstly, when optimizing constraint-related variables, the DE/current-to-$p$best/1 strategy is used, which can improve the convergence of the population and help it find promising feasible regions more easily. Afterward, the DE/better/1 strategy is employed to optimize constraint-independent variables with the aim of enhancing population diversity. This allows the population to further explore within the identified feasible regions, finding areas with better quality objectives, and ultimately covering the entire CPF. Unlike other algorithms, DVCEA does not utilize the relationship between objectives and constraints to guide the evolution of the population, so it can concentrate on searching for promising feasible areas and exploring deeply within feasible areas without the interference of complex constraints. Therefore, DVCEA can achieve superior performance on SDC test problems with complex constraints. As for the SDC6 test problem, the overall constraint terrain is relatively flat, and the proportion of feasible areas is also very small, so the population is easy to fail to find feasible areas or falls into local areas. The reason why DVCEA performs less well than IMTCMO may be because the DVCEA's population only finds part of feasible areas, so it cannot search all CPF fragments. The diversity of IMTCMO is relatively strong because of its powerful search mechanism, so it can find more feasible regions, and thus perform better than the proposed algorithm. For SDC15, the constraint violation degree near CPF is small, while the opposite is true near UPF. If the information of objectives is used too much, the population will not be able to fully explore the region near CPF, such as PPS. On the contrary, algorithms that utilize both objectives and constraints information simultaneously are more likely to find CPF, such as NSGAIITOR, ShiP, CMOEA_MS, and BiCo. But overall, DVCEA still exhibits superior performance on most problems. Moreover, DVCEA also achieved satisfactory performance on the ZXH_CF test problems with simple constraints, indicating that guiding population evolution through the relationship between decision variables and constraints does not lead to performance degradation on general problems.

Table S-4 of the supplementary file shows the specific HV results. According to the HV values, the performance of DVCEA is significantly better than that of NSGAIIToR, ShiP, PPS, CMOEA_MS, IMTCMO, and BiCo on 22, 25, 31, 22, 22, and 16 problems respectively, while it is significantly worse than them only on 1, 1, 0, 3, 0, and 3 problems respectively. In addition, NSGAIIToR, ShiP, PPS, and CMOEA_MS cannot obtain 100% FSR respectively on 3, 7, 2, and 4 problems, while IMTCMO, BiCo, and DVCEA cannot achieve 100% FSR only on 1 problem. Based on the above results and analysis, DVCEA shows its superiority in solving problems with complex and simple constraints, and DVCEA is very competitive compared to the comparison algorithms.

C. Statistical Analysis and Results

Table S-12 shows the Wilcoxon test results of DVCEA and the other six algorithms at the significance level of 0.05. The experiments are conducted by KEEL software [50]. If the difference between R^+ and R^- is greater, it indicates a more significant disparity between the algorithms. If the P-value is below 0.05, it indicates a significant difference between the algorithms at a significance level of $\alpha \leq 0.05$. It can be observed that all R^+ values surpass R^- values, and in every instance, DVCEA exhibits significant distinctions compared to other algorithms, thereby demonstrating its superiority.

Fig. 9 illustrates the average rankings of IGD^+ and HV results obtained by seven algorithms across 31 problems by the Friedman test. From the ranking results, it can be observed that DVCEA achieved the lowest ranking value, indicating that DVCEA exhibits superior performance compared to other algorithms. Furthermore, for a more intuitive visualization of the performance of each algorithm, four test problems, SDC2, SDC3, SDC4, and ZXH_CF15, are selected to demonstrate the results of the algorithms, that is final population distribution maps of the median IGD^+ achieved by these algorithms on these four problems. The specific analysis and distribution diagrams are shown in Section III in the supplementary file.

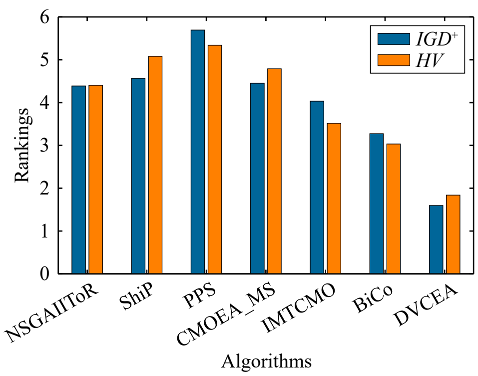


Fig. 9. Rankings of DVCEA and compared algorithms on average IGD^+ and HV values.

D. Experimental Verification of DVCEA

In this section, some parameters and main mechanisms of DVCEA are verified, including the selection of parameters K and PE, the decision variables classification method, and the proposed offspring generation strategies.

1) Parameters Analysis: When decision variables are classified according to their effect on constraints, there are two important parameters: the number of selected solutions K and the number of perturbations PE on each variable, which determine the quality of the decision variables classification. If the parameters are set too small, the decision variables may not be correctly classified. However, if the parameters are set too large, although it can ensure the correctness of the classification, it will lead to excessive consumption of computing resources, resulting in a smaller proportion of computing resources used for subsequent optimization and a poorer quality of optimization. In this paper, K and PE are set to 5 and 4, respectively. In order to select the appropriate parameter settings, different combinations of parameters are set: $K = 5$, $PE = 2$; $K = 5$, $PE = 6$; $K = 5$, $PE = 11$; $K = 10$, $PE = 2$; $K = 10$, $PE = 4$; $K = 10$, $PE = 6$; and $K = 10$, $PE = 11$. The variant algorithms are all named in the format of DVCEA_ K_PE.

Table S-5 and S-6 respectively present the mean IGD^+ and HV values for different combinations of parameters K and PE. It can be observed that similar performance is achieved for all parameter combinations except DVCEA_5_2 and DVCEA_10_2. The main reason for this is that the decision variables are perturbed too few times in these two variant algorithms, as a result, the decision variables are not classified correctly. Whereas in the other parameter combinations, most of the decision variables can be classified correctly, the difference is that the classification in the different combinations uses different computational resources, resulting in different computational resources that they can use in the subsequent optimization. Before the optimization is completed, the variant algorithms can converge to the vicinity of CPF. Therefore, these different parameter combinations achieve similar results. However, in general, when $K = 4$ and $PE = 5$, the computational resources used in the classification are the least, and therefore the computational resources used in the subsequent optimization process are also the most, thus showing relatively better results than other parameter combinations.

In fact, the ideal state is to use the least amount of computing resources to achieve the maximum classification effect. Here, the SDC test problem suit is used as an example to show the classification situations of different parameter combinations. The specific classification accuracy for various parameter combinations on the 15 SDC test problems are listed in Fig. S-5 in the supplementary file. It can be observed that when the number of perturbations PE on each dimension is 2, the classification accuracy is the lowest, indicating that only two perturbations are not enough to classify the decision variables correctly, and more perturbations are needed. When PE is 4, most of the decision variables can be correctly classified. As the number of individuals selected and the number of perturbations increases, the classification accuracy also increases, but at the same time, the computing resources consumed are gradually increasing. Therefore, in order that more computational resources can be used for subsequent optimization, a compromise solution is chosen in this paper, i.e., $K = 5$ and $PE = 4$.

2) The Verification of Decision Variables Classification Method: The classification of decision variables is a critical step in preparing for subsequent optimization and has a direct impact on the outcome of the optimization process. Therefore, the decision variables classification method is a crucial mechanism in DVCEA. To validate the effectiveness of the method, several variant experiments are established: The first variant, referred to as DVCEA_NCP, does not classify decision variables, but directly optimizes them using strategy DE/current-to-$p$best/1. In the second variant, the decision variables are not classify, and the strategy DE/better/1 is used to optimize them, called DVCEA_NCB. The third variant, DVCEA_NC_PB, does not classify decision variables, and two evolutionary strategies have the same probability of being called to optimize all variables. In the fourth variant, the decision variables are randomly classified, and each variable has a probability of 0.5 to be classified as either constraint-related variables or constraint-independent variables, expressed as DVCEA_RA; For the last variant, represented as DVCEA_ LM, the decision variables are classified using the classification method in the large-scale algorithm LMEA [40]. In LMEA, decision variables are categorized into convergence variables and diversity variables, considering their impact on the objective function, while DVCEA_LM uses the classification methods in LMEA to classify variables according to their

TABLE II
THE IGD^+ VALIDATION EXPERIMENTAL RESULTS OF DECISION VARIABLES CLASSIFICATION METHOD

Problem	DVCEA_NCP	DVCEA_NCB	DVCEA_NC_PB	DVCEA_RA	DVCEA_LM	DVCEA
SDC1	1.0754e+1 (3.85e−1) −	1.0186e+1 (1.38e+0) −	1.0648e+1 (9.24e−1) =	1.0553e+1 (1.14e+0) −	9.8559e+0 (5.56e−1) =	1.0148e+1 (1.41e−1)
SDC2	7.4969e−1 (1.38e−2) −	3.9138e−1 (4.10e−2) =	7.3550e−1 (4.25e−2) −	4.1167e−1 (6.69e−2) −	5.0268e−1 (5.18e−2) −	3.8472e−1 (1.43e−1)
SDC3	1.5688e+0 (1.35e−1) −	1.3807e+0 (6.55e−2) =	1.4782e+0 (3.41e−2) −	1.4518e+0 (1.04e−1) −	1.3613e+0 (6.87e−2) =	1.3433e+0 (6.78e−2)
SDC4	1.5461e+0 (8.18e−2) −	5.2235e+0 (5.95e+0) =	1.5517e+0 (1.03e−1) −	1.4445e+0 (9.06e−2) =	9.3274e+0 (1.60e+1) −	1.4434e+0 (5.69e−2)
SDC5	1.3476e+3 (1.72e+2) −	1.4152e+3 (1.69e+2) −	1.1072e+3 (1.80e+2) =	1.3743e+3 (2.19e+2) −	1.2132e+3 (1.09e+2) =	1.1389e+3 (2.05e+2)
SDC6	7.1565e+2 (1.15e+3) =	1.0645e+3 (9.73e+2) −	2.4789e+2 (3.55e+2) =	4.0912e+2 (4.54e+2) −	NaN (92%) −	3.4798e+2 (3.57e+2)
SDC7	5.0895e+0 (6.75e−1) −	2.9869e+0 (2.07e−1) −	2.6591e+0 (3.58e−1) −	3.4842e+0 (3.35e−1) −	3.1972e+0 (9.55e−1) −	2.1953e+0 (2.83e−1)
SDC8	5.1244e+1 (3.02e+1) −	6.3417e+1 (2.18e+1) −	3.5620e+1 (1.76e+1) −	5.3153e+1 (1.93e+1) −	4.4218e+1 (3.07e+1) −	7.8429e+0 (8.68e+0)
SDC9	1.3299e+3 (2.29e+2) −	3.2512e+2 (7.85e+1) +	6.9624e+2 (2.07e+2) −	4.2042e+2 (8.99e+1) =	NaN (92%) −	4.5855e+2 (9.38e+1)
SDC10	1.0247e+1 (2.14e−1) −	1.0292e+1 (1.42e+0) =	1.0417e+1 (6.10e−1) −	1.0440e+1 (1.44e+0) −	1.0080e+1 (4.68e−1) =	9.9617e+0 (2.66e−1)
SDC11	4.0576e+2 (1.21e+2) =	5.5454e+2 (1.02e+2) −	4.4059e+2 (1.21e+2) =	4.3275e+2 (1.42e+2) =	5.7880e+2 (1.87e+2) −	4.0670e+2 (8.43e+1)
SDC12	4.5510e+1 (2.58e+1) −	9.1932e+1 (2.25e+1) −	3.6085e+1 (1.78e+1) −	7.9762e+1 (3.05e+1) −	6.9319e+1 (4.26e+1) −	8.3027e+0 (7.46e+0)
SDC13	1.3544e+1 (6.55e−1) −	NaN (80%) −	1.3365e+1 (9.47e−1) =	1.2471e+1 (4.08e−1) =	1.2956e+1 (6.02e−1) −	1.2645e+1 (7.26e−1)
SDC14	NaN (52%) −	NaN (12%) −	NaN (32%) −	NaN (32%) −	NaN (40%) −	NaN (80%)
SDC15	1.1721e+1 (5.80e−1) −	9.8782e+0 (2.02e−1) =	1.0589e+1 (1.88e−1) −	1.0256e+1 (1.47e−1) −	1.0069e+1 (5.68e−1) =	1.0083e+1 (4.64e−2)
ZXH_CF1	5.9955e−2 (3.09e−3) −	4.1182e−2 (8.83e−4) =	4.6048e−2 (1.04e−3) −	4.1460e−2 (2.79e−3) =	4.1398e−2 (1.01e−3) =	4.0332e−2 (1.21e−3)
ZXH_CF2	1.2751e−1 (1.20e−1) =	2.5046e−1 (5.24e−1) =	4.7944e−1 (5.43e−1) −	3.9996e−1 (6.02e−1) −	NaN (92%) −	1.6561e−1 (2.76e−1)
ZXH_CF3	1.0688e−1 (4.31e−3) −	5.8277e−2 (2.58e−3) =	7.4611e−2 (3.81e−3) −	6.6323e−2 (6.82e−3) −	5.7906e−2 (5.07e−3) =	5.8004e−2 (2.98e−3)
ZXH_CF4	6.2183e−2 (2.03e−3) −	4.2254e−2 (3.21e−3) −	3.7810e−2 (1.71e−3) −	4.5515e−2 (8.38e−3) −	8.0700e−2 (7.08e−2) −	3.3352e−2 (1.70e−3)
ZXH_CF5	3.6119e−1 (3.03e−1) −	4.1885e−1 (5.43e−1) −	3.3759e−1 (4.39e−1) −	1.3842e−1 (1.73e−1) −	2.6258e−1 (4.16e−1) −	7.1258e−2 (9.07e−2)
ZXH_CF6	3.6539e−2 (1.44e−3) −	2.2568e−2 (9.44e−4) +	2.6826e−2 (1.26e−3) −	2.4406e−2 (3.05e−3) =	1.9342e−2 (6.09e−4) +	2.3469e−2 (5.39e−4)
ZXH_CF7	3.4079e−2 (1.49e−3) −	7.2694e−2 (7.77e−2) −	2.3600e−2 (9.50e−4) −	3.0988e−2 (4.69e−3) −	3.1029e−2 (4.07e−2) −	2.1831e−2 (7.05e−4)
ZXH_CF8	6.2219e−2 (2.15e−3) −	3.6976e−2 (1.19e−3) =	4.0771e−2 (1.59e−3) −	4.4259e−2 (4.37e−3) −	2.3285e−2 (7.48e−4) +	3.5995e−2 (1.31e−3)
ZXH_CF9	2.7678e−2 (1.84e−3) −	1.6412e−2 (7.07e−4) −	1.6384e−2 (6.10e−4) −	1.6737e−2 (1.57e−3) −	1.1165e−2 (6.22e−4) +	1.4022e−2 (5.62e−4)
ZXH_CF10	2.2852e−2 (8.06e−4) −	1.6220e−2 (1.35e−3) −	2.5149e−2 (3.75e−2) −	2.9412e−2 (4.28e−2) −	2.3540e−2 (3.92e−2) −	1.2401e−2 (2.86e−4)
ZXH_CF11	1.3069e−2 (4.64e−4) −	9.5267e−3 (2.68e−4) −	9.4456e−3 (2.39e−4) −	9.7753e−3 (2.86e−4) −	8.9757e−3 (2.47e−4) =	9.0386e−3 (2.57e−4)
ZXH_CF12	2.5252e−1 (3.02e−1) =	1.3493e−1 (2.33e−1) =	5.3003e−2 (7.29e−2) =	3.1207e−1 (3.22e−1) −	NaN (92%) −	1.6745e−1 (2.01e−1)
ZXH_CF13	2.0413e−3 (1.22e−4) −	2.0289e−3 (1.96e−4) −	2.4917e−2 (7.53e−2) −	1.9655e−2 (5.41e−2) −	1.8254e−2 (5.43e−2) −	1.1719e−3 (5.62e−5)
ZXH_CF14	6.3397e−3 (2.28e−4) −	3.0967e−3 (1.68e−4) −	2.2680e−3 (5.79e−5) −	3.5497e−3 (7.56e−4) −	2.1845e−3 (3.47e−5) =	2.1886e−3 (5.52e−5)
ZXH_CF15	1.0144e−1 (1.97e−1) =	6.3110e−2 (1.46e−1) =	1.0086e−1 (2.21e−1) =	1.0415e−1 (1.87e−1) −	NaN (80%) −	6.3026e−2 (1.07e−1)
ZXH_CF16	1.4855e−3 (9.46e−5) −	1.1451e−3 (4.04e−5) =	1.1505e−3 (4.18e−5) =	1.1906e−3 (4.25e−5) −	1.1271e−3 (2.52e−5) =	1.1188e−3 (4.88e−5)
+/−/=	0/26/5	2/17/12	0/23/8	0/25/6	3/16/12	

influence on constraints. The IGD^+ and HV experimental results are given in Tables II and Table S-7, respectively.

From the IGD^+ experimental results, it can be observed that DVCEA is significantly superior to its variant algorithms on 26, 17, 23, 25, and 16 problems, respectively, while it is significantly inferior to the variant algorithms on only 0, 2 0, 0, and 3 problems. For the first three variants, the decision variables are not classified, that is, all variables are optimized together. In this way, the difficulty of optimization is increased due to the increase of decision variables, resulting in poor optimization results. In addition, DVCEA_NCP uses DE/current-to-$p$best/1 to optimize all decision variables, DVCEA_NCB uses DE/better/1 to optimize decision variables, while DVCEA_NC_PB mixes the two strategies, but they did not achieve the desired effect, mainly because there is no specific optimization for specific decision variables, which further demonstrates the effectiveness of classifying decision variables. In DVCEA_RA, decision variables are randomly classified, and the difficulty of optimization is reduced, but decision variables with the same characteristics are not divided together, resulting in the optimization strategy is no longer targeted, as a result, the optimization effect is greatly reduced. As for DVCEA_LM, the classification method in LMEA is employed. Specifically, each variable is disturbed several times, after which multiple new individuals generated by the disturbance will be fitted, then a clustering operation will be performed according to the angle between the fitted line and the normal of the hyperplane. The decision variables are then clustered into convergence variables and diversity variables. However, in the SDC test problems, the constraints are notably complex and non-linear, leading to considerable errors during fitting and clustering. Consequently, there is a higher likelihood of misclassification during the process, as indicated by the average classification accuracy in Fig. S-5 of the supplementary file. Additionally, it utilizes a substantial amount of computational resources during the decision variables classification process. As a consequence, the subsequent optimization experiences a decline in performance, and the variant algorithm's efficiency is inferior to that of DVCEA. However, for ZXH_CF test problems with less complex features, it shows good results and the efficiency is verified. In the proposed algorithm, multiple perturbations are applied to each dimension, and decision variables are categorized directly based on the extent of constraint changes. This approach is more favorable for observing the effect of decision variables on constraints, resulting in a more accurate classification.

Similarly, according to the HV results, DVCEA significantly outperforms the variant algorithms on 25, 17, 19, 19, and 15 problems, while it is significantly inferior to them on only 0, 1, 1, 0, and 3 problems, respectively. The main reason why the performance of variant algorithms is not as excellent as DVCEA is twofold: first, DVCEA can separate variables with different characteristics, and secondly, it can perform targeted optimization on two different types of variables. Specifically, constraint-related variables have a large influence on constraints, and the optimization of such variables can quickly help the population locate the feasible regions. Moreover, DE/current-to-$p$best/1 strategy can assist individuals to learn better information, thus guiding the evolution of the population towards a more promising region and accelerating the convergence of the population. When the constraint independent variables are optimized, the change of constraints is minimal. Therefore, DE/better/1 strategy is designed to help increase the diversity of the population. In this strategy, paired individuals are randomly selected to learn from each other, resulting in more learning directions being generated, so that the population can explore more extensively near the searched feasible region to help it better find CPF. The variant DVCEA_NCP uses DE/current-to-$p$best/1 to optimize all variables, although this can increase the convergence of the population, once the population falls into the local optimal, it is difficult for the population to escape, so it does not perform as well as DVCEA on most problems. Variant DVCEA_NCB uses DE/better/1 to optimize all variables, the diversity of the population is increased, and more potential areas can be searched. Therefore, it can also achieve excellent results on some complex problems, such as SDC9 and SDC15, but it cannot guarantee the convergence of the population, so its performance cannot exceed that of DVCEA on most problems. Variants DVCEA_NC_PB and DVCEA_RA use random optimization and random grouping mechanisms, and it is evident that they both lack targeted optimization strategies, resulting in significantly worse performance than DVCEA. Based on the aforementioned analysis, the proposed decision variables classification method is effective and capable of accurately classifying the majority of problems.

3) Verification of the Effectiveness of the Combination of Offspring Generation Strategies: The excellent offspring generation strategies can facilitate the evolution of population towards promising regions. In DVCEA, two strategies, DE/current-to-$p$best/1 and DE/better/1, are used to optimize constraint-related variables and constraint-independent variables, respectively, with the aim of ensuring both convergence and diversity of the population. To investigate the effectiveness of the combination of offspring generation strategies, several comparative experiments are configured: For the first variant, both types of variables are optimized using DE/current-to-$p$best/1, referred to as DVCEA_$p$best. Conversely, for the second variant, DE/better/1 is employed to optimize these two types of variables, labeled as DVCEA_better. The third variant employs DE/better/1 to optimize constraint-related variables, while DE/current-to-$p$best/1 is used to optimize constraint-independent variables, called DVCEA_BR_PI. In contrast to the optimization sequence of DVCEA, the last variant, DVCEA_RE, first optimizes constraint-independent variables using DE/better/1 and then optimizes constraint-related variables using DE/current-to-$p$best/1. The detailed IGD^+ results are shown in Table S-8.

The experimental results reveal that DVCEA significantly outperforms the four variants on 19, 15, 25, and 1 test problems, respectively. However, it exhibits significantly worse performance than them on only 1, 4, 1, and 0 problems. In the first variant, since DE/current-to-$p$best/1 is employed to optimize both decision variables, the convergence ability of population is improved, but the diversity will decrease, so it cannot escape the local optimal region in the later stage of evolution, thus its performance in most problems is worse than DVCEA. For the second variant, DE/better/1 is used to optimize both variables. In fact, because two individuals are randomly selected in the population for comparison, which prevents all individuals from learning in the same direction and getting stuck in a local region. In addition, the learning strategy designed promotes superior individuals to evolve in a better direction, while poorer individuals absorb valuable information without compromising diversity, so it takes both convergence and diversity into account to some extent, and exhibits excellent results on some problems. However, its convergence performance is not as good as DE/current-to-$p$best/1, so the performance of variant is inferior to DVCEA. In variant DVCEA_BR_PI, DE/better/1 is used to optimize constraint-related variables, while DE/current-to-$p$best/1 is used to optimize constraint-independent variables. Because strategy DE/current-to-$p$best/1 is more biased towards convergence, while DE/better/1 is more biased towards diversity, this approach cannot achieve the goal of quickly converging by optimizing constraint-related variables and increasing population diversity by optimizing constraint-independent variables. Therefore, it performs significantly worse than the proposed algorithm on most problems. The variant DVCEA_RE changes the optimization order of the two types of decision variables, and it shows a similar performance to DVCEA, indicating that the optimization order does not affect the quality of optimization. In fact, from the perspective of single-generation optimization, the optimization order of the two types of variables is relatively obvious, but from the perspective of the entire optimization process, the optimization is carried out alternately, and there is no obvious optimization order. Therefore, the performance of the variant algorithm is comparable to that of DVCEA.

The specific HV values are listed in Table S-9 in the supplementary file. From the experimental results, the performance of DVCEA significantly exceeds that of variant algorithms DVCEA_$p$best, DVCEA_better, and DVCEA_BR_PI on 20, 15, and 26 problems, respectively. Conversely, DVCEA performs significantly worse than them on 1, 4, and 1 problems, respectively. Compared to variant algorithm DVCEA_RE, the two algorithms exhibit comparable performance.

Based on the aforementioned results and analysis, the combination of offspring generation strategies has proven to be effective. This demonstrates the capability to concurrently address population convergence and diversity, guiding the population to evolve in more promising directions.

E. Application to Real-World Problems

To validate the performance of the algorithms in practical scenarios, 20 real-world problems in literature [20], RWCMOP1-RWCMOP20, all of which are derived from mechanical design problems, are selected. Furthermore, since the true CPFs for these problems are unknown, the best non-dominated solutions among solutions found by all algorithms including DVCEA are extracted to form the reference solution set to calculate the IGD^+ indicator for assessing the performance of all algorithms. In addition, the specific experimental results are presented in Table III.

From the results, the performance of DVCEA is significantly better than NSGAIIToR, ShiP, PPS, CMOEA_MS, IMTCMO, and BiCo on 18, 19, 18, 19, 8, and 9 problems, while it is significantly worse than that of the comparison algorithms only on 0, 0, 2, 0, 2, and 3 problems. In fact, most real-world problems contain complex constraints. For example, the objectives and constraints of the car side impact design problem (RWCMOP8) show opposite trends, if too much information about the objectives is utilized, the population will explore in the opposite direction of CPF, and there will no longer be sufficient resources to explore the optimal region. While DVCEA will help the population explore promising feasible areas, as well as areas between UPF and CPF, to prevent the population from consuming too much computing resources in UPF exploration. In addition, there is no clear relationship between the objectives and constraints of the front rail design problem (RWCMOP18), and excessive use of objectives information can lead to the population falling into local optima. But DVCEA directly considers the relationship between variables and constraints while ignoring the relationship between objectives and constraints, Therefore, it can help the population locate promising feasible regions by optimizing constraint-related variables, and further explore this region by optimizing constraint-independent variables, making it easier to search for CPF. Furthermore, it can be observed that both IMTCMO and BiCo also exhibit excellent performance on some practical problems, which are algorithms based on two-population, indicating the effectiveness of such a method. By cooperating between two populations with different purposes, the algorithm can achieve favorable outcomes. However, for most problems with complex relationships between objectives and constraints, they still cannot achieve satisfactory performance, which is mainly due to their lack of targeted optimization strategies. On the contrary, DVCEA performs different optimization operations on decision variables with different characteristics, so it can achieve excellent results on most practical problems with complex constraints, which also proves its ability to solve real-world problems. In addition, Fig. 10 shows the average IGD^+ values of several variant algorithms on the pressure vessel design problem (RWCMOP1). From the results, it can be seen that DVCEA achieves the smallest average value. This is mainly because the first three variants do not classify variables and all variables are optimized simultaneously. The last one randomly classifies variables, but due to the inaccuracy of classification, the designed optimization strategies can not be targeted to optimize the corresponding variables. In both cases, populations tend to fall into the local optimal. While DVCEA can help the population evolve towards promising regions by optimizing different variables using different optimization strategies. Firstly, DVCEA uses the DE/current-to-$p$best/1 strategy to optimize constraint-related variables, thereby accelerating population convergence and helping the population move towards promising feasible regions. Then, the DE/better/1 strategy is employed to optimize constraint-independent variables to improve the diversity of the population and assist the population to better search for CPF. However, due to the low dimensionality of the RWCMOP1 problem, there is no significant difference in the results between these variants and DVCEA. But overall, DVCEA can still achieve the best performance on practical problems compared to peer algorithms and variant algorithms.

V. Conclusion

In this paper, an optimization algorithm based on the classification of decision variables is designed for solving CMOPs, known as DVCEA. In the proposed approach, all decision variables are categorized into two types based on their relationship with constraints: constraint-related variables and constraint-independent variables. Subsequently, these two types of variables are optimized separately to reduce the complexity of the search. Additionally, two offspring generation strategies with different functions are designed and applied to the

TABLE III
AVERAGE IGD^+ VALUES ON 20 REAL-WORLD PROBLEMS

	The methods based on single population and single stage		The methods based on two-stage		The methods based on two-population		The proposed algorithm
Problem	NSGAIIToR	ShiP	PPS	CMOEA_MS	IMTCMO	BiCo	DVCEA
RWCMOP1	1.3858e+05 (6.78e+03) −	1.3106e+05 (4.67e+03) −	1.6904e+05 (8.33e+03) −	1.8755e+07 (9.45e+06) −	1.0247e+05 (2.75e+03) =	1.0120e+05 (1.83e+03) +	1.0548e+05 (4.25e+03)
RWCMOP2	5.8507e−02 (1.98e−03) −	NaN (0%) −	7.5235e−02 (2.88e−02) −	2.4327e+01 (5.09e+01) −	4.1250e−02 (9.82e−04) −	2.0624e+00 (6.32e+00) −	4.1015e−02 (1.13e−03)
RWCMOP3	2.6289e+02 (1.38e+01) −	2.6538e+02 (1.17e+01) −	3.1232e+03 (1.30e+03) −	7.6182e+03 (5.22e+02) −	2.3832e+02 (1.22e+01) =	2.3907e+02 (9.90e+00) =	2.4541e+02 (2.16e+01)
RWCMOP4	1.5531e−01 (7.61e−03) −	1.5229e−01 (8.16e−03) −	3.2946e−01 (2.97e−02) −	NaN (43.2%) −	8.7188e−02 (1.25e−03) =	9.7246e−02 (2.87e−02) −	9.3292e−02 (1.21e−02)
RWCMOP5	2.0236e−02 (4.82e−03) −	1.8765e−02 (6.68e−04) −	1.8552e−02 (1.41e−03) −	7.8665e−02 (9.16e−03) −	1.2598e−02 (2.69e−04) −	1.2295e−02 (2.21e−04) =	1.2261e−02 (1.52e−04)
RWCMOP6	9.5604e+00 (4.92e−01) −	9.5009e+00 (3.91e−01) −	2.0812e+01 (4.78e+00) −	1.0808e+01 (4.03e−01) −	8.5148e+00 (2.46e−01) −	8.1779e+00 (2.24e−01) =	8.1554e+00 (2.44e−01)
RWCMOP7	3.7039e−02 (1.59e−03) −	3.5535e−02 (8.15e−04) −	3.1115e−02 (4.63e−04) −	2.9883e−02 (1.21e−03) −	2.5888e−02 (5.10e−04) =	2.6519e−02 (5.60e−04) −	2.5774e−02 (5.43e−04)
RWCMOP8	3.4341e−01 (1.35e−02) −	3.3133e−01 (1.37e−02) −	5.4884e−01 (7.56e−02) −	3.8302e−01 (1.30e−02) −	2.7407e−01 (4.10e−03) =	3.5486e−01 (1.88e−02) −	2.7056e−01 (3.23e−03)
RWCMOP9	4.9987e+00 (1.77e−01) −	5.0935e+00 (1.99e−01) −	1.8381e+01 (3.62e+00) −	8.1667e+00 (5.85e−01) −	4.2545e+00 (1.37e−01) =	4.1409e+00 (8.84e−02) =	4.2310e+00 (1.08e−01)
RWCMOP10	3.7949e−01 (2.17e−02) −	4.0469e−01 (3.65e−02) −	1.2477e+00 (3.20e−01) −	1.1008e+01 (2.12e−01) −	2.8836e−01 (9.60e−03) =	3.2215e−01 (7.77e−03) −	2.8444e−01 (9.96e−03)
RWCMOP11	4.3045e+04 (2.30e+03) −	4.1999e+04 (3.12e+03) =	3.4716e+04 (2.30e+03) +	5.6093e+04 (9.02e+03) −	4.8975e+04 (2.60e+03) −	5.4259e+04 (3.08e+03) −	4.0189e+04 (1.06e+03)
RWCMOP12	2.9624e+00 (1.47e−01) −	2.9610e+00 (1.38e−01) −	1.3508e+01 (1.52e+01) −	NaN (32.1%) −	1.7687e+00 (9.16e−02) −	1.7203e+00 (3.63e−02) =	1.7146e+00 (2.39e−02)
RWCMOP13	6.5316e+01 (4.31e+00) −	6.3956e+01 (3.42e+00) −	9.8430e+01 (4.27e+00) −	NaN (11.9%) −	4.8990e+01 (8.51e−01) −	4.8134e+01 (1.31e+00) =	4.7963e+01 (1.28e+00)
RWCMOP14	4.6993e−03 (1.85e−04) −	4.5127e−03 (1.69e−04) −	3.4542e−02 (2.22e−02) −	NaN (48.4%) −	2.9654e−03 (5.04e−05) =	2.9350e−03 (5.31e−05) =	2.9249e−03 (7.48e−05)
RWCMOP15	4.6572e+01 (1.08e+01) =	NaN (40%) −	3.3155e+02 (1.10e+02) −	4.0507e+01 (2.85e+01) =	2.1706e+01 (4.31e+00) +	2.2526e+01 (3.12e+01) +	3.7682e+01 (1.24e+01)
RWCMOP16	9.3361e−03 (4.79e−04) −	9.1621e−03 (3.19e−04) −	8.8457e−03 (5.69e−04) −	9.2158e−02 (9.64e−03) −	6.7435e−03 (2.02e−04) =	6.4800e−03 (8.83e−05) +	6.7484e−03 (2.06e−04)
RWCMOP17	1.9611e+19 (1.05e+17) =	NaN (92%) −	5.3975e+18 (5.37e+18) +	1.9678e+19 (3.55e+09) −	1.9678e+19 (4.39e+11) −	1.9678e+19 (2.76e+09) −	1.9678e+19 (9.06e+14)
RWCMOP18	7.6564e−04 (4.09e−05) −	7.7802e−04 (3.23e−05) −	8.8030e−04 (3.86e−05) −	5.9962e−04 (6.73e−06) −	5.9425e−04 (1.19e−05) −	5.9485e−04 (1.24e−05) =	5.9034e−04 (8.38e−06)
RWCMOP19	7.2951e+03 (2.53e+03) −	NaN (0%) −	2.7120e+03 (3.21e+02) −	7.5538e+03 (3.57e+03) −	6.7220e+02 (9.77e+01) =	1.7233e+03 (1.41e+03) −	1.5332e+03 (1.91e+03)
RWCMOP20	NaN (60%) −	4.7062e+02 (2.37e+02) −	6.3661e+02 (1.14e+02) −	1.8817e+03 (3.90e+02) −	9.2338e+01 (3.88e+01) +	4.5709e+02 (2.69e+02) −	4.1092e+02 (5.61e+02)
+/−/=	0/18/2	0/19/1	2/18/0	0/19/1	2/8/10	3/9/8	

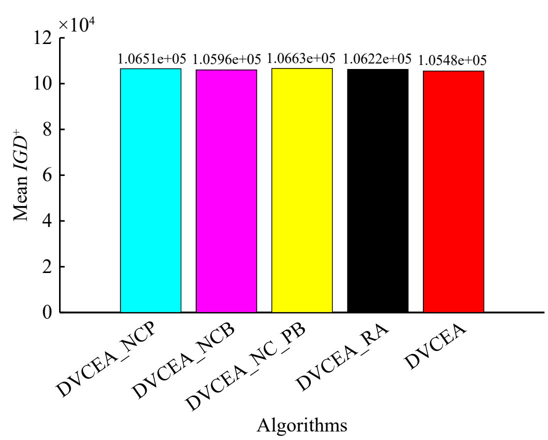


Fig. 10. Average IGD^+ results of DVCEA and its variant algorithms on the pressure vessel design problem.

two types of decision variables respectively to achieve the purpose of accelerating the population convergence and maintaining the population diversity. The proposed algorithm is compared with six algorithms in three different categories on 31 test problems and 20 practical problems, and the experimental results confirm the effectiveness of the algorithm.

However, the proposed algorithm still has some limitations, for example: 1) The coupling relationship between variables is not considered, and the algorithm is easy to falls into local optimal when the variables with coupling relationship are not optimized simultaneously; 2) The change of the influence of variables on constraints in the evolutionary process is not considered. With the evolution of the population, the influence of variables on constraints may change at different stages, and the initial classification results may not be applicable to the subsequent evolution. In future work, the DVCEA will be further improved to solve the above limitations, for example, in order to solve the first limitation, variables can be grouped based on coupling relationships, and the relationship between different groups and constraints can be determined. Then these groups can be divided into constraint-related variable groups and constraint-independent variable groups, and then optimized respectively. For the second limitation, the relationship between variables and constraints can be re-evaluated every certain number of evolutionary generations, and then optimized based on the new grouping. Furthermore, DVCEA can be developed to solve more complex CMOPs, such as large-scale constrained multi-objective optimization problems (LSCMOPs). In LSCMOPs, the search space expands dramatically due to the increase in the number of decision variables, so it is difficult for the algorithm to explore the whole search space. If the decision variables are divided into two categories according to their relationship with constraints and then optimized separately, it is equivalent to exploring the population in two low-dimensional subspaces respectively, which is more conducive to in-depth exploration of the search space. In addition, decision variables can be further refined based on their relationship with the objectives, and the influence of the variables on the objective values can be judged on the basis of the designed classification method, so that the variables can be divided into more categories, so as to make the optimization process more targeted and solve the LSCMOPs better.

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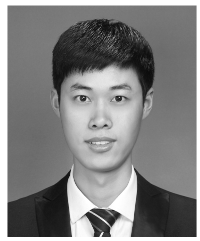

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